Digital Logic
Day 5

uarch.space

Sandbox

Click the + icon to place components down.

Place something down, press Ctrl + S, and reload the page. It should still be there!

Your progress is also autosaved every 10 seconds.

Ctrl + S (save)

Saves your progress to a .json file. Keep this file save!

Ctrl + O (open)

Opens a .json file to load progress. Transfer progress across computers!

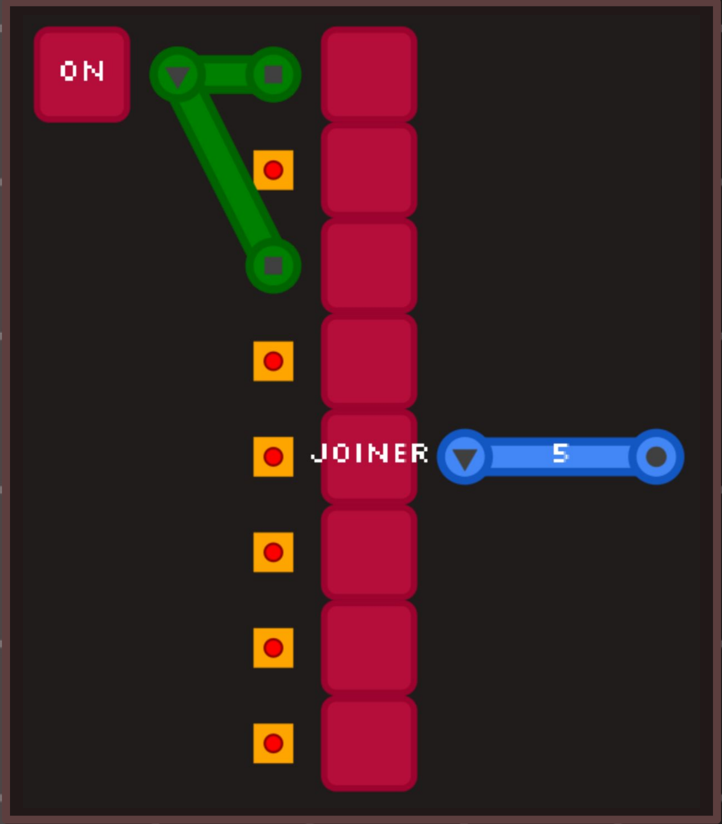
Bundles!

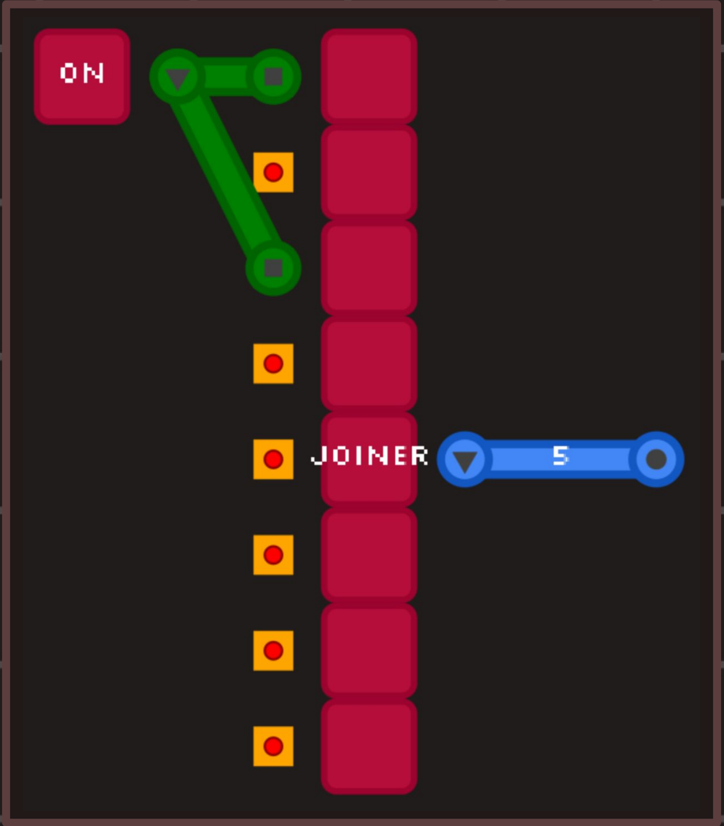
Find the “Joiner” component in the [+] menu.

On the output tile ▶, right click and drag. The wire should be blue.

Left click = normal wire

Right click = bundle wire

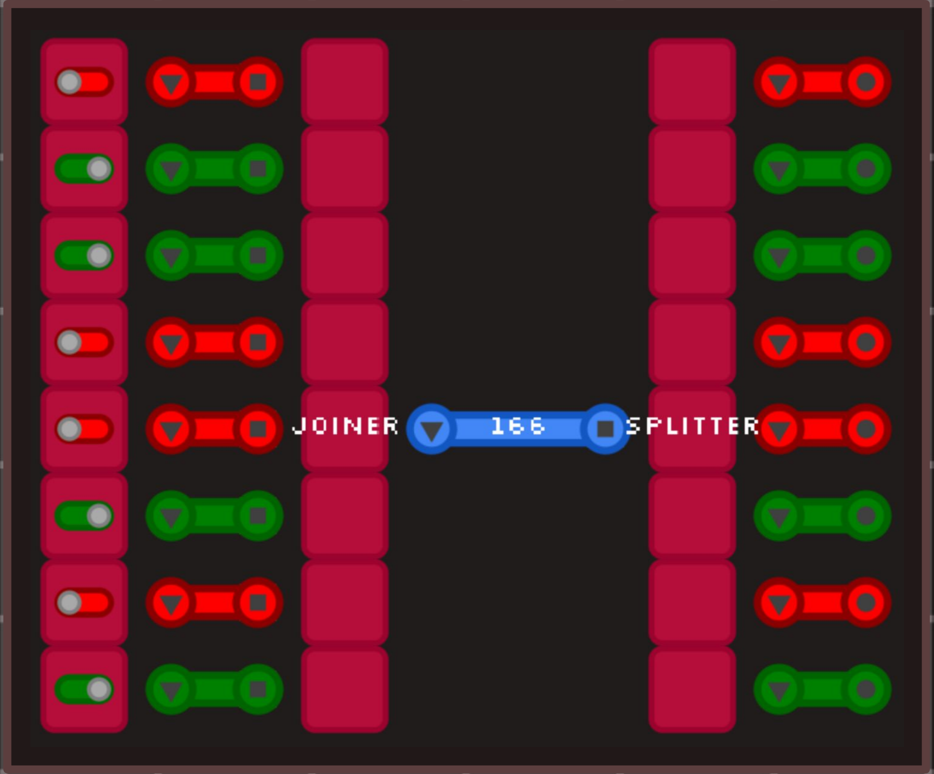




Bundles!

The bundle wire is a bundle of 8 wires joined together.

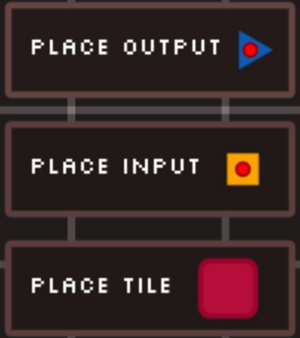
Why do you think the number 5 is displayed on the wire?



The “Splitter” does the opposite of the joiner and gives us the signals that make up a bundled wire.

Making Components

Click the icon to create a new component.



We will make a "ADD8" component - an 8 bit adder.

Important!

Width: 24

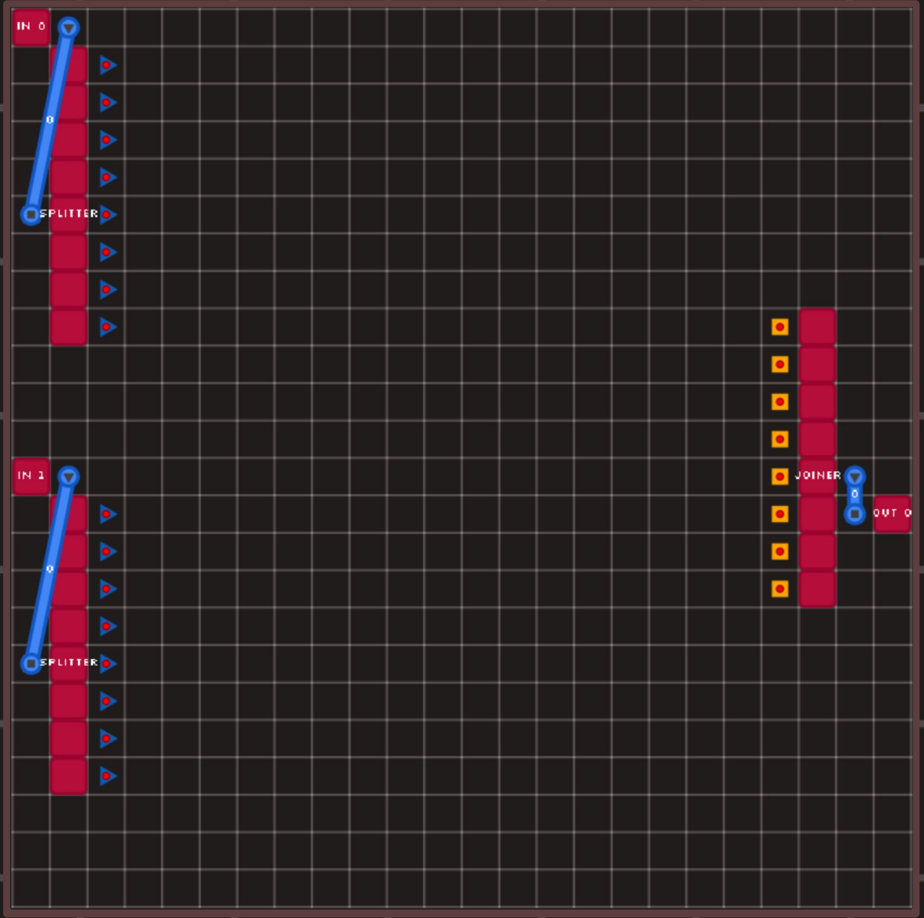
Height: 24

Click and drag tiles from the left side to make your schematic.

Add 8

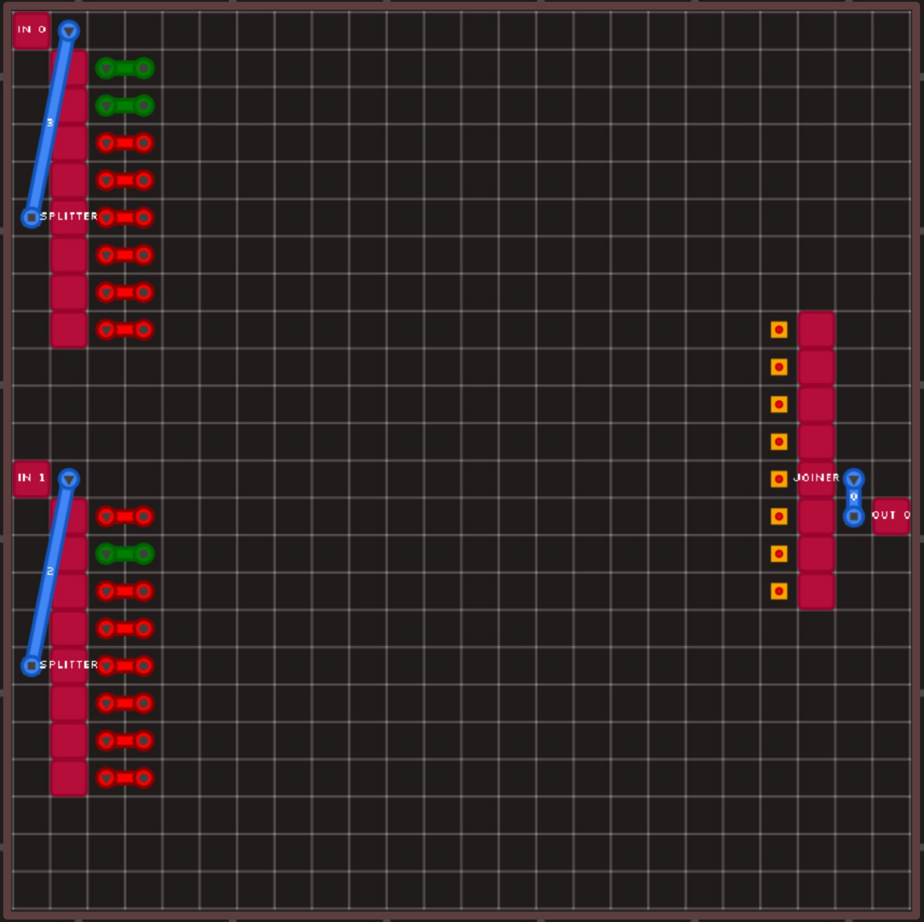
Setup your component like this:

The two binary numbers to add on the left side, and the right side is the output



0000 0011 (3)

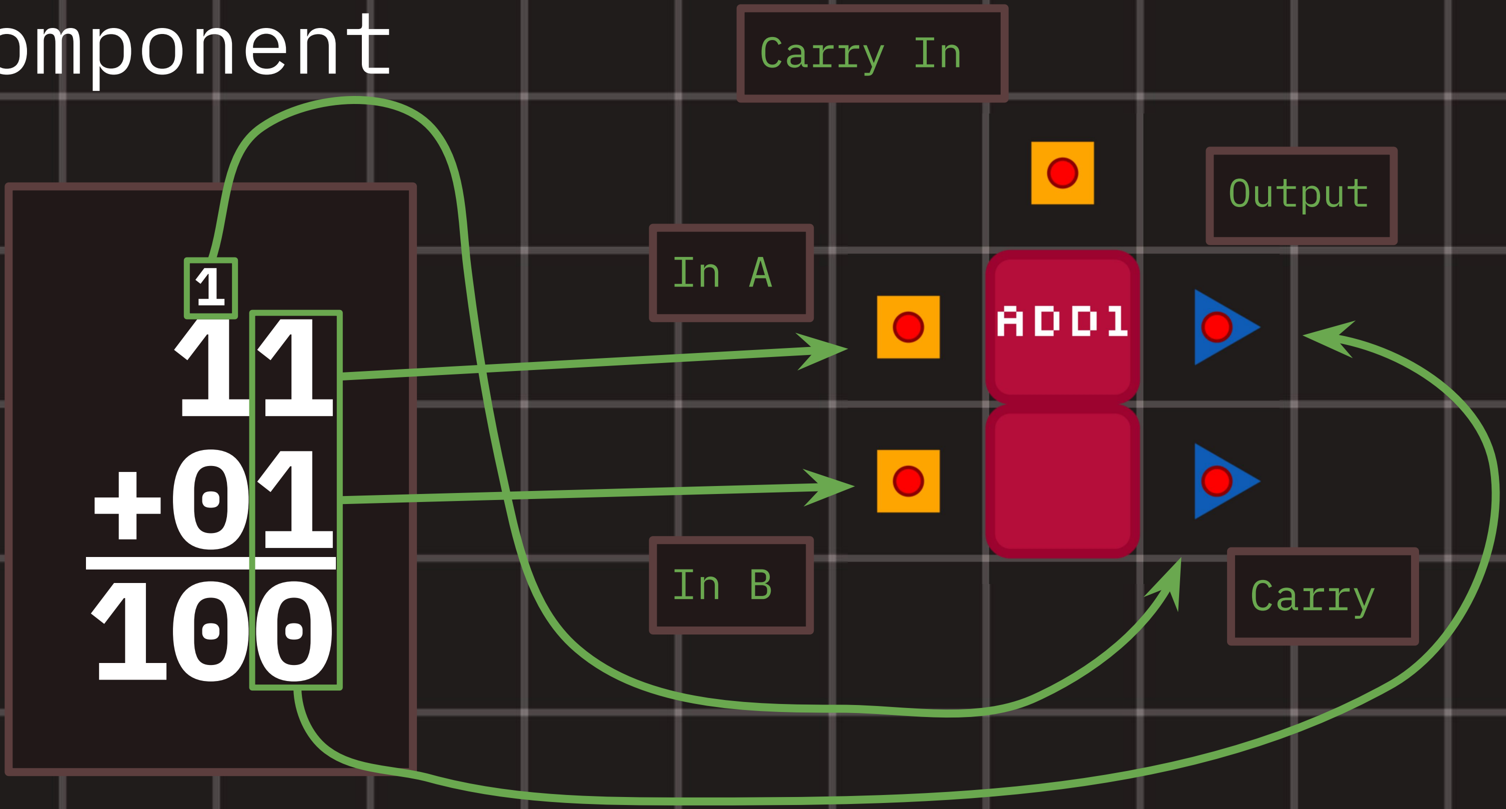
0000 0010 (2)



Intended output:

0000 0101 (5)

Full Adder Component



The First Bit

The first bit (the first column) doesn't have a carry in, so nothing connects to it.

We take the least significant bit from the two binary inputs and add them together.

You'll add the rest of the bits - make sure to use the carry!